



An Efficient Optimization-Based Approach to Determine Benchmark Failure Probability in Slope Reliability

Nhu Son Doan¹

Accepted: 11 December 2024

© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2025

Abstract

This study introduces an efficient approach to evaluate benchmark failure probabilities in slope reliability. Particle swarm optimization (PSO) and the firefly algorithm (FA) are compared to determine the most effective strategy for slope stability analysis. It was found that, in FA, the movements of fireflies are adjusted based on the performance of other fireflies, resulting in a computational burden of approximately five times greater than that of PSO. Consequently, PSO was integrated into Monte Carlo simulations for system reliability assessments. Three case studies on system slope reliability were rigorously analyzed, with soil properties modeled as either random variables or random fields. The results obtained using the proposed method showed strong agreement with those from the traditional method and existing literature, affirming its accuracy. Notably, a fivefold reduction in the number of failure surfaces requiring evaluation was achieved when soil properties were treated as random variables. Furthermore, by accounting for spatial variability in soil—thereby increasing problem complexity—the analysis was accelerated by a factor of ten, highlighting the remarkable efficiency of the proposed approach.

Keywords Slope reliability · Monte Carlo simulation · Limit equilibrium method · Particle swarm optimization · Firefly algorithm

1 Introduction

Safety factors reflecting the stability of slopes are commonly determined by either the limit equilibrium or finite element analyses (Javankhoshdel et al. 2021; Duncan et al. 2014; Doan 2023). Thus, deterministic analyses of slope stability are implicit procedures. In turn, probabilistic analyses, which define the limit state equations

✉ Nhu Son Doan
vanson.ctt@vmaru.edu.vn

¹ Faculty of Civil Engineering, Vietnam Maritime University, 484 Lach Tray Str., Haiphong City, Vietnam 180000

based on the deterministic safety factors, become implicit processes (Ji and Low 2012). Moreover, slope reliability generally involves multiple failure modes; hence, it is vital to select appropriate methods to ensure an accurate estimation of the system failure probability (Ji and Low 2012). Monte Carlo simulation (MCS) is commonly chosen for slope reliability since it can address the difficulty of implicit and multimodal problems (Doan 2023; Van et al. 2023). In MCSs, populations of uncertain variables are first generated, and the safety factors for each sample will be evaluated to establish the overall population. MCS offers several significant advantages. First, MCS involves a repetitive process requiring numerous deterministic calculations, making it straightforward and accessible for most engineers without deep knowledge of probabilistic methods. Second, MCS is highly effective at handling implicit processes, often outperforming FOSM and FORM. (Doan 2023; Halder and Mahadevan 2000; Dinh et al. 2024). Theoretically, a slope can fail along numerous potential slip surfaces, making it difficult to determine the overall failure probability across all surfaces. In practice, a set of representative failure surfaces is commonly utilized to evaluate the system failure probability in slope stability analysis. Nevertheless, the most notable drawback of MCS is its computational inefficiency. This inefficiency stems from the repetitive nature of MCS, with computing time increasing not only with the size of the simulation but also with the number of representative failure modes. For instance, completing a single cycle of MCS may require several hours (Ji and Low 2012; Doan et al. 2024a).

In slope stability analysis, thousands of candidate failure surfaces are often examined to identify the most critical failure mode (Doan et al. 2024b; Zhang et al. 2013). Each candidate surface represents a potential failure mode, with the most critical being the one exhibiting the lowest safety factor. In probabilistic analyses, each candidate failure surface is considered an uncertain event, and the slope is deemed unstable if any of these failure events occur. Consequently, slope reliability is evaluated as a series of system-level problems. (Ji and Low 2012). The entry and exit technique (EET) is the most widely used method in the limit equilibrium approach for generating candidate failure surfaces (Doan et al. 2024b; Zhang et al. 2013). This technique assumes multiple entry points along the slope crest, while exit points are placed near the toe. Center points are then established along a perpendicular line between each pair of entry and exit points, creating thousands of potential failure surfaces. Consequently, safety factors must be evaluated repeatedly across these surfaces to identify the most critical failure mode in deterministic analysis (Xiao et al. 2017). Furthermore, to ensure comprehensive exploration, the boundaries for entry, exit, and center points must be sufficiently broad, with densely distributed surveyed points to accurately detect the most critical failure surface. However, increasing the number of surveyed points and expanding investigation ranges can produce a large set of candidate surfaces, adversely affecting computational efficiency. In the existing literature, the EET is frequently integrated with MCS, referred to as basic MCS in this study, to assess system failure probability in slope reliability. Results from basic MCS are regarded as exact solutions and serve to validate newly proposed methods. However, the extensive number of candidate surfaces generated by EET can substantially increase the computing time and effort required for MCS, which typically involves thousands to hundreds of thousands of simulations.

Two approximation approaches can be employed to enhance computational efficiency. The first involves variance reduction techniques that reduce the size of the MCS, thereby improving efficiency (Halder and Mahadevan 2000). The most widely used approaches for slope reliability are importance sampling (Ching et al. 2009) and subset simulation (Yang et al. 2018). These techniques can significantly enhance computational performance by allowing smaller MCS sizes. Alternatively, surrogate models, such as the traditional response surface method (Ji and Low 2012) or more recent machine learning models (Doan et al. 2024a; Ahmad et al. 2024; Kang et al. 2015), can be developed to replace the original implicit problems. Using explicit surrogate models can significantly improve computational efficiency without reducing the MCS size (Doan et al. 2024a; Kang et al. 2015). Each approach requires specialized knowledge to ensure accurate implementation. For instance, variance reduction techniques demand a solid understanding of probabilistic calculations to correctly select instrumental density functions (Yi et al. 2015), while traditional response surface methods may not yield accurate results if the performance function is highly nonlinear (Yi et al. 2015). Therefore, benchmark failure probabilities remain essential for validating these approximation methods. In the literature, extensive studies have focused on proposing approximation methods; however, there has been limited attention on developing approaches to obtain benchmark solutions.

This study proposes effective methods to derive benchmark solutions for slope reliability while improving computational efficiency. Since these benchmarks depend on performing MCSs for initial analyses, the primary focus is on developing facile methods for deterministic slope stability calculations. To achieve this, an optimization approach is introduced to minimize the extensive calculations required by the EET. This proposed method facilitates the rapid identification of the most critical failure surface across a wide range of potential scenarios without significantly increasing the number of candidate surfaces. Subsequently, the optimization approach is integrated into MCSs to derive benchmark failure probabilities for slope reliability.

The remainder of this study is structured as follows: Section 2 covers deterministic slope stability calculations and probabilistic analyses using MCSs for slope reliability. It introduces the optimization concept, outlines the algorithms employed for deterministic calculations, and details the integration of the optimization approach into probabilistic analysis. Section 3 presents three numerical examples of slope reliability, considering both random variables and random fields, to demonstrate the efficiency and accuracy of the proposed method. Finally, Section 4 concludes the study.

2 Methods

This section outlines the deterministic calculations for slope stability and the probabilistic analyses of slope reliability. Section 2.1 summarizes the conventional approach for slope stability analyses, which employs entry and exit techniques to generate a set of potential failure surfaces. As discussed later, numerous candidate surfaces must be evaluated to accurately detect the most critical failure surface. Therefore, Section 2.2

proposes an optimization to improve slope stability assessment. Two nature-inspired optimization algorithms, particle swarm optimization (PSO) and firefly algorithm (FA), are applied to slope stability assessments. Finally, Section 2.3 details the integration of the optimization approach with the MCS for slope reliability.

2.1 Conventional Approach for the Slope Stability Assessment

Considering a slope presented in Fig. 1a, the EET can be implemented to generate potential failure surfaces, as shown in Fig. 1b. In deterministic calculations, the safety factors for all candidate surfaces will be calculated, and the minimum value determines the safety factor of the slope. Several slice methods, such as Fellenius, Bishop, Spencer, and Morgenstern-Price, can be adopted to evaluate the safety factor for each failure surface (Duncan et al. 2014; Doan 2023). Since the number of unknown variables exceeds the equilibrium conditions, these methods are distinguished by their assumptions regarding the failure shapes, the interaction forces between slices, and the satisfaction of equilibrium for moments, forces, or both (Duncan et al. 2014). In practice, Bishop's simplified method (BSM) is commonly recommended in design codes for its accuracy and efficiency (AASHTO. LRFD 2017; MLIT 2020), and it is therefore adopted in this study. Notably, circular failure surfaces are assumed in BSM.

Considering any potential failure surface shown in Fig. 2(a), the safety factor for this surface can be estimated using Eq. (1a) with the BSM. The overall safety factor for slope stability (FS_{Sys}) is determined by Eq. (1b), which implies that the system factor is represented by the minimum safety factor detected among all considered surfaces.

$$FS = \frac{\sum \left[\frac{c\Delta/\cos\alpha + (W - u\Delta/\cos\alpha)\tan\phi}{\cos\alpha + (\sin\alpha\tan\phi)/FS} \right]}{\sum W\sin\alpha} \quad (1a)$$

$$FS_{Sys} = \min\{FS\} \quad (1b)$$

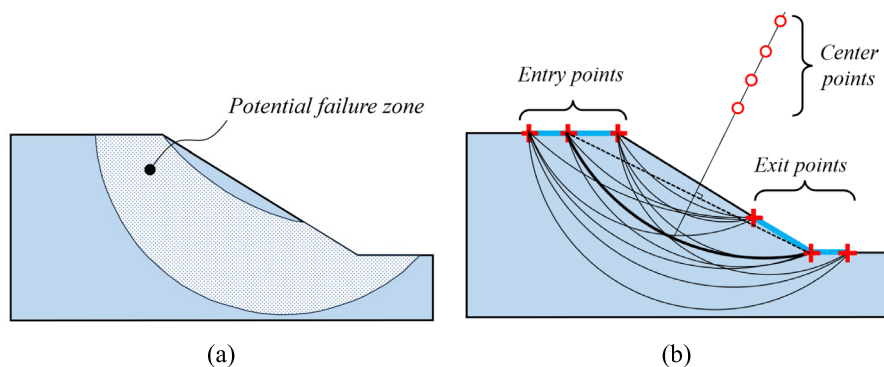


Fig. 1 Illustrative failure modes: **a** potential failure zone and **b** failure surfaces generated by EET

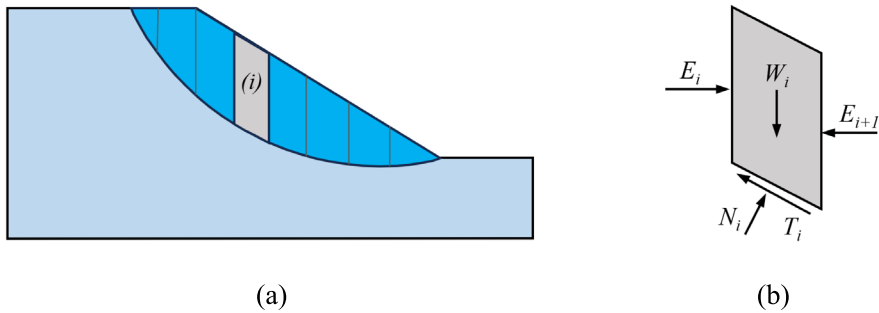


Fig. 2 Illustration of a failure mode: **a** failure mass and **b** free body of a slice

In Eq. (1a), ϕ and c are the friction angle and cohesion of soils at the base of the slice, respectively. Δl and α correspond to the base length of the slice and the angle of the baseline with the horizontal direction. W is the weight of the slice. The forces acting on a free slice are illustrated in Fig. 2b, and the sum in Eq. (1a) is taken for all slices divided. Notably, the BSM assumes that the shear forces between slices are equal, and the normal force N in Fig. 2b is assessed by taking the force equilibrium in the vertical direction. Extensive explanations about the BSM and limit equilibrium methods can be referred to in previous works, e.g., (Duncan et al. 2014).

Figure 1 and Eq. (1) indicate that, to ensure an accurate prediction of FS_{sys} , the survey of the potential failure zone must be sufficiently comprehensive. Generally, the possible failure zone is defined by the entry and exit ranges and the selected center points. Furthermore, to achieve an accurate FS_{sys} , the number of points within each range must be densely generated, necessitating the evaluation of thousands of potential surfaces. Handling thousands of potential surfaces is feasible in deterministic analysis, as stability is assessed only once. However, it becomes significantly more challenging in direct MCSs, where this vast number of candidate surfaces must be recalculated tens to hundreds of thousands of times to complete a single simulation. Consequently, the following subsection proposes an optimization approach to mitigate the computational burden of evaluating all candidate failure surfaces in MCSs.

2.2 Optimization-Based Approach for Slope Stability Assessment

This subsection presents an optimization-based approach for evaluating slope stability to overcome the inefficiency of EET discussed previously. The optimization approach aims to find the most critical failure surface without considering many candidate failure surfaces presented in Section 2.1.

Considering the slope in Fig. 1 again, a failure surface can be generated by selecting a random entry point within the entry range and a random center point inside the potential center zone, as shown in Fig. 3. The safety factor associated with the surface generated can be assessed by Eq. (1). The optimization focuses on finding the most critical failure surface with the minimum safety factor, which can be

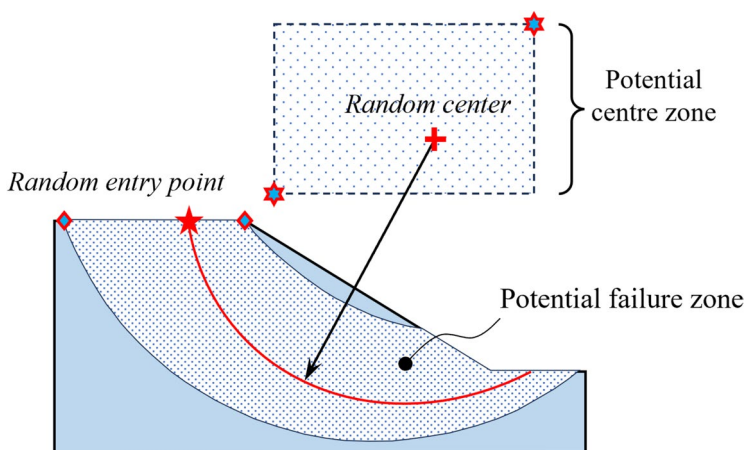


Fig. 3 Illustration of the proposed optimization approach

mathematically written using Eq. (2). In Eq. (2), $\mathbf{x}_E$ denotes the coordinate of an entry point randomly created within an entry range surveyed ($\mathbf{R}_E$). Likewise, $\mathbf{x}_C$ depicts the coordinates of any center point inside the potential center zone ($\mathbf{R}_C$). $\mathbf{s}$ is a vector including all soil and geometry properties relating to the underlying surface. Theoretically, Eq. (2) can be solved by any optimization method; however, the optimization may be more efficient than the conventional procedure if the number of trial surfaces explored is fewer than that generated by EET.

$$FS_{Sys} = \min_{\substack{\mathbf{x}_E \in \mathbf{R}_E \\ \mathbf{x}_C \in \mathbf{R}_C}} \{FS(\mathbf{x}_E, \mathbf{x}_C, \mathbf{s})\} \quad (2)$$

The particle swarm optimization and the firefly algorithm optimization are used in this work to search for the most critical failure surface. Extensive insight into the PSO can be referred to previous works such as (Djaneye-Boundjou et al. 2009; Rukhaiyar et al. 2018). Alternatively, FA was well presented in the earlier work (Yang 2009) and applied to most engineering fields (Li et al. 2022). The procedures of PSO and FA are briefly summarized as follows.

PSO is a population-based optimization method that mimics the social behavior of groups, allowing a swarm of particles to work together to find optimal solutions in the search space. At the first iteration, the positions of the particles, representing initial guesses for the optimization variables, are randomly generated within the search domain. The objective functions are then evaluated for each position of particles. Afterward, velocities ($\mathbf{v}$) and positions ($\mathbf{x}$) of particles in the next iteration (iteration $it+1$) are modified by considering the global best (g^{Best}) and the particle best (p^{Best}) found in the preceding iteration (iteration it), as shown in Eq. (3). $\mathbf{rand}_p$ and $\mathbf{rand}_G$ in Eq. (3a) denote random vectors depicting amounts of randomness for the current iteration. The optimization variable $\mathbf{x}$ can be defined by combining $\mathbf{x}_E$ and $\mathbf{x}_C$ in Eq. (2). Acceleration coefficients c_1 and c_2 can be taken as 2.05, as recommended

in Djaneye-Boundjou et al. (2009). The inertia weight w can be turned as a constant (e.g., 0.9) or gradually decreased by iterations (Banks et al. 2007). Generally, a higher inertia weight is beneficial during the exploitation stage, facilitating global search, while smaller inertia weights are more effective for local searching and exploration. In this work, the inertia weight is iteratively reduced from 0.9 to 0.4 to balance global and local search capabilities. The maximum number of iterations can define the termination criterion of the optimization process. Aside from turning the weight and acceleration coefficients, selecting reasonable numbers of particles in the swarm and the maximum iterations are crucial for achieving efficient optimizations. Therefore, this study will evaluate the appropriate number of particles and iteration limits for PSO to ensure an effective integration with MCSs. The procedure of PSO is summarized in Fig. 4a.

$$v_{it+1}^P = wv_{it}^P + c_1 \text{rand}_P(p_{it}^{\text{Best}} - x_{it}^P) + c_2 \text{rand}_G(g_{it}^{\text{Best}} - x_{it}^P) \quad (3a)$$

$$x_{it+1}^P = x_{it}^P + v_{it+1}^P \quad (3b)$$

Based on the idealization of the flashing features generated by fireflies, FA aims to mimic the behaviors of fireflies during the searches for mating partners or to

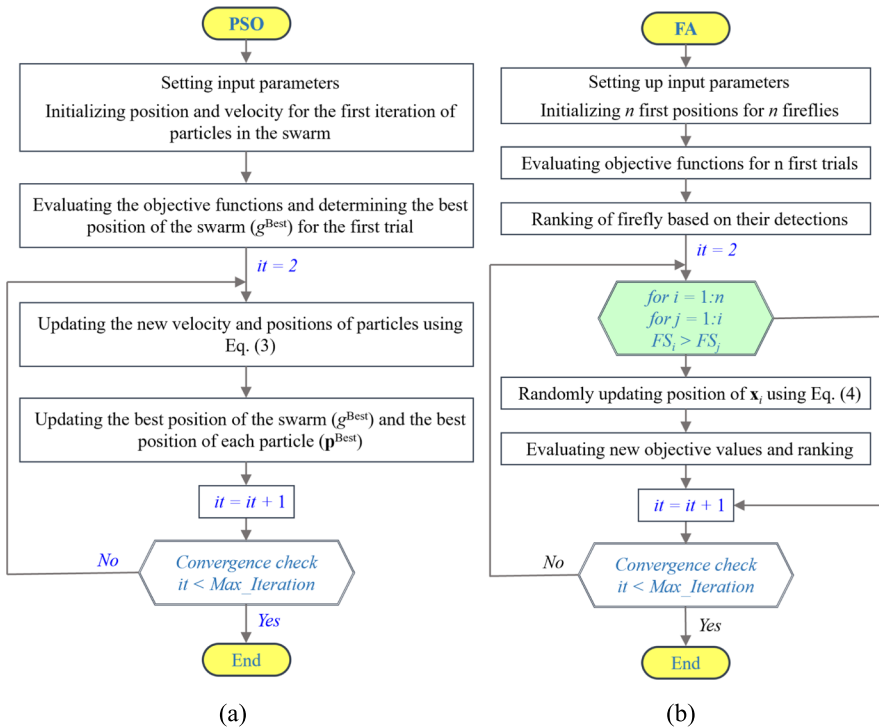


Fig. 4 Flowcharts of the optimizations: **a** particle swarm optimization and **b** firefly algorithm

attract potential prey. FA was reported to be superior to PSO when dealing with multimodal optimization problems (Yang 2009; Yang 2011).

The movement of a firefly ($\mathbf{x}_i$) is attracted by the brighter flashes of another firefly ($\mathbf{x}_j$), and it can be mathematically formulated by Eq. (4a). The Cartesian distance r between the two fireflies can be determined based on Eq. (4b).

$$x_i(it+1) = x_i(it) + \beta_0 e^{-\gamma r_{ij}^2} [x_j(it) - x_i(it)] + \alpha \epsilon_i(it) \quad (4a)$$

$$r_{ij}^2 = \sum_1^{\text{size of } x} (x_j - x_i)^2 \quad (4b)$$

In Eq. (4a), β_0 is the attractiveness of a light source at a zero distance ($r=0$). The exponential function implies that the attractiveness of a light source is conversely related to the distance between the two fireflies. γ is the light absorption coefficient, which can be taken from 0.01 to 100 in most engineering problems (Yang 2009). The second term in Eq. (4a) reflects the attraction of the firefly $\mathbf{x}_i$ to the brighter firefly $\mathbf{x}_j$. The third term in Eq. (4a) accounts for a random amount of the movement to the brighter source, reflected by the randomization parameter α (between 0 and 1) and a random vector ϵ_i generated by Gaussian, uniform, or other distributions (Yang 2011). The procedure of FA is summarized in Fig. 4b. Notably, the PSO routine operates with a single trial iteration loop (i.e., it). In contrast, FA consists of three loops: one for trial iteration, akin to PSO, and two inner loops that update firefly positions by comparing their brightness. It is essential to highlight that these inner loops require additional evaluations of the performance function, specifically the safety factor estimations utilized in this study. The effects of these inner loops will be investigated in Section 3.

In slope stability analyses, the accuracy and efficiency of traditional methods using EET depend on the defined search domains and how effectively points are refined within these domains. In contrast, the optimization-based approaches are controlled by factors such as the number of particles or fireflies and the maximum number of optimization iterations. Notably, these quantities can be represented by the number of failure surfaces that must be investigated in each method. Thus, the necessary number of failure modes examined in each process is evaluated. Comparisons on the number of failure surfaces will be discussed in each example in Section 3.

2.3 Methods of MCS-Based Slope Reliability

MCS is a robust method for dealing with implicit processes such as slope or breakwaters' foundation stabilities (Doan 2023; Boudiaf et al. 2019). In slope reliability, the limit state equation g is commonly defined based on the system safety factor, as shown in Eq. (5a). Equation (5a) implies that the failure event occurs whenever FS_{sys} is lower than unity (i.e., g is smaller than zero). Thus, the system failure probability P_f , estimated by MCS-based reliability analysis, can be written as Eq. (5b). In the Equation, N_{MCS} is the number of realizations

created by MCS. The symbol $I[\bullet]$ represents an indicator function, which returns a value of 1 when the inside condition is met and 0 when the inside condition is not met. Thus, P_f is the ratio between the number of failure events observed in the simulations and the size of the MCS. The accuracy of MCS is influenced by its sample size. The coefficient of variation of the failure probability (COV_p), defined in Eq. (6), is frequently used to evaluate this accuracy (Haldar and Mahadevan 2000). Equation (6) indicates that the estimated failure probability approaches zero as the sample size of the MCS becomes infinite (Doan 2023). In practice, an appropriate size for the MCS is determined based on an acceptable threshold for the COV_p , typically around 30% for geotechnical problems (Ching et al. 2009).

$$g(x) = FS_{sys} - 1 \quad (5a)$$

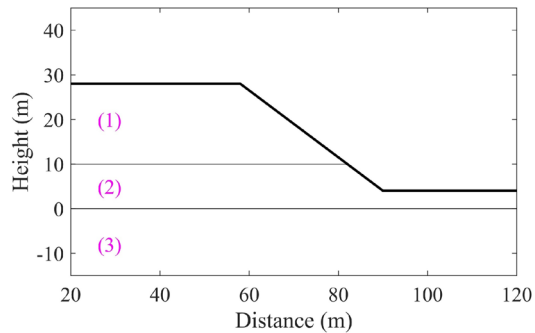
$$P_f = P(g(x) < 0) = \frac{\sum_1^{N_{MCS}} I[g(x) < 0]}{N_{MCS}} \quad (5b)$$

$$COV_p \approx \frac{1}{\sqrt{N_{MCS} P_f}} \quad (6)$$

It is important to note that the system safety factor is calculated based on the original problems through the optimization process (Eqs. (1) and (2)). Therefore, the proposed optimization-based MCS remains an implicit process. However, the number of candidate surfaces is expected to decrease significantly, leading to a more accurate and efficient procedure, as demonstrated in Section 3.

3 Numerical Examples

This study analyzes three slopes to demonstrate the efficiency and accuracy of the proposed approach. For each case, soil profiles and the statistical properties of uncertainties are presented, along with the search domains for deterministic calculations. Reliability analyses are conducted using the proposed method to evaluate failure probabilities. In the first two examples, soil properties are modeled as random variables, while in the third, they are simulated as random fields to account for spatial variability. The basic MCS method is also performed for comparison, and the results are compared against those available in the literature. Notably, a smaller COV_p indicates a more accurate estimate. Moreover, evaluating the safety factor for each potential failure surface is both implicit and computationally intensive. Thus, the number of possible failure surfaces required in the reliability analyses serves as a critical metric to assess the performance of MCSs.

Fig. 5 Cross-section of Example 1**Table 1** Soil properties and statistics in Example 1

Item	Layer 1			Layer 2		
	γ (kN/m ³)	ϕ (°)	c (kPa)	γ (kN/m ³)	ϕ (°)	c (kPa)
Mean	19	—	120	19	—	160
COV	—	—	0.3	—	—	0.3
Distribution	—	—	Normal	—	—	Normal

3.1 Slope Example 1

In Example 1, a slope consisting of two clays located on a hard layer, as shown in Fig. 5, is investigated. This example was taken from previous works (Ji and Low 2012; Kang et al. 2015), and the soil properties of clays are listed in Table 1. In the deterministic calculation, a safety factor of 1.992 was reported in Kang et al. (2015). Moreover, a basic MCS with 20,000 samples was performed, and a failure probability of 0.405% was reported in the previous work (Ji and Low 2012). However, the previous studies did not mention the number of potential failures surfaces investigated.

In this work, 4000 candidates (consisting of 10 entry points, 25 exit points, and 16 centers), as shown in Fig. 6a, are generated using the EET to represent the potential failure zone. The most critical safety factor estimated from these candidates is 1.993, as reported in Fig. 6a. When using the PSO, the search domains for the entry and center points are depicted in Fig. 6b by diamond and star markers, respectively. The effects of selecting particle numbers and the maximum iterations will be presented later in Fig. 8. When 25 particles and 30 iterations are used, the trial failure surfaces and corresponding center points are recorded in Fig. 6b. The most critical failure surface with a safety factor of 1.993 is found after examining 750 failure surfaces, as reported in the figure. The results in Fig. 6 demonstrate that the optimization approach outperforms the conventional EET by requiring significantly fewer investigated failure surfaces, reducing approximately six times.

The two optimization algorithms are implemented using the same combination of 25 particles and 30 iterations, and their performances are compared in Fig. 7 for ten different runs. The color numbers in the figure denote the safety factors

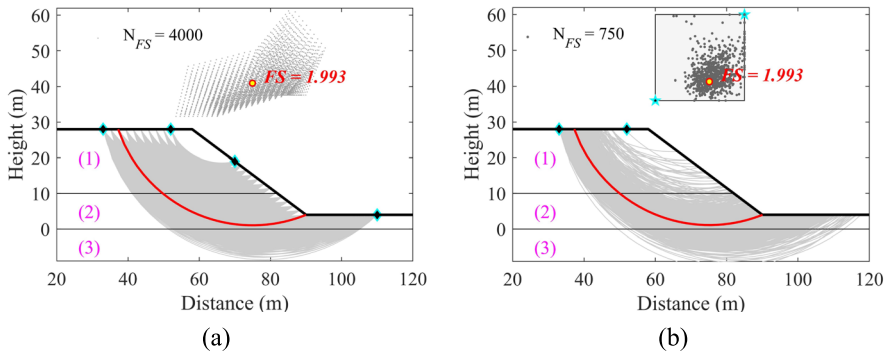


Fig. 6 Deterministic results of Example 1: **a** using EET and **b** using PSO approach

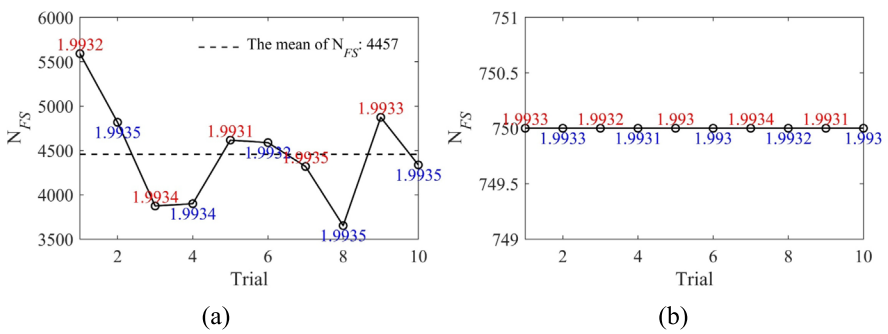


Fig. 7 Numbers of the failure surfaces required in the optimizations for Example 1: **a** using FA and **b** using PSO

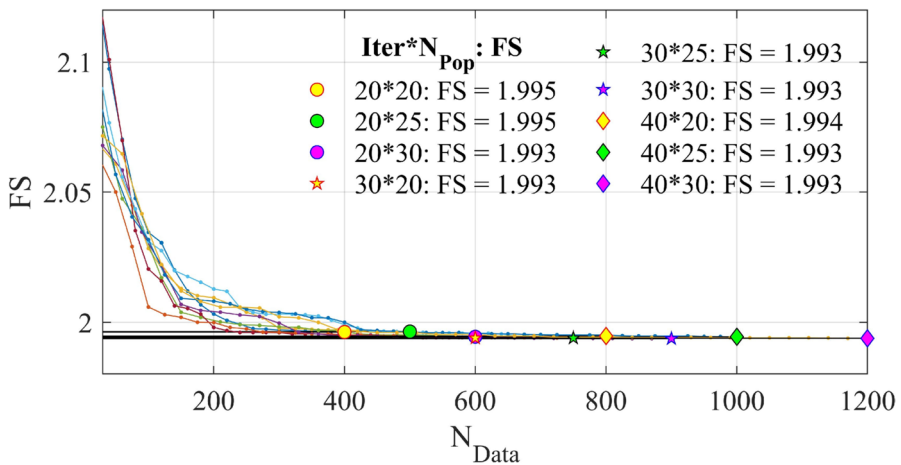


Fig. 8 Comparisons for various combinations of particle number and iteration for Example 1

estimated for each trial. The vertical axes indicate the necessary numbers of failure surfaces (N_{FS}) until the most critical failure mode is identified for each run. The figure shows that consistent safety factors are determined using both optimization methods. Nevertheless, the number of the failure surfaces needed to investigate varies by the trial runs when using FA, while it is the same when using PSO (750 surfaces, i.e., equal to the production of numbers of particles and the maximum iterations). The reason is the movements of particles in PSO are based on their best positions and the best position of the swarm in the preceding iteration, and no interaction between particles is required (see Eq. (3)). On the other hand, the movements of fireflies are influenced by the performances of other fireflies, as shown by the second term in Eq. (4a) and the inner loop (green box) in Fig. 4. This interaction results in a variable number of calculations across different trial runs. Figure 7a shows that the number of surfaces investigated using the FA varies by iteration, with a mean of 4457. This is higher than the conventional EET and significantly greater than the PSO, which is approximately six times lower. Consequently, PSO is further implemented for the probabilistic analysis.

Different combinations of the number of particles in the swarm and the limit of iterations in the optimization are examined. In Fig. 8, comparisons are made to select the appropriate combination for Example 1. Notably, ten different runs are executed, and the average results for the ten trials are summarized in Fig. 8. Figures 6b and 8 show that the combination of 25 particles and 30 iterations provides good performance, and this combination is utilized in the probabilistic analysis. The results of the probabilistic analyses using EET and the optimization are reported in Table 2. The fourth column of the table provides the total count of failure surfaces examined in the MCSs. These values are calculated by multiplying the MCS size by the number of failure surfaces examined for each method. The results are also compared with those reported in the literature. Table 2 demonstrates that this work achieves better results. Notably, the small COV of P_f indicates a higher resolution in the P_f estimate. Interestingly, the number of failure surfaces investigated (as shown in the fourth column) is reduced by approximately fivefold using the optimization approach compared to the conventional entry and exit technique. Consequently, computing time can be reduced by approximately fivefold, allowing for faster execution of the MCS using the procedure introduced in this work.

Table 2 Comparisons of probabilistic results for Example 1

Method	P_f (%)	COV_P (%)	Number of failure surfaces in MCS	Reference
EET-based for critical failure surface and MCS (50,000 samples)	0.426	6.90	2×10^8 (50,000 $\times$ 4000)	This study
PSO-based for critical failure surface and MCS (50,000 samples)	0.426	6.90	3.75×10^7 (50,000 $\times$ 750)	This study
Basic MCS with 20,000 samples	0.405	11.11	–	Kang et al. 2015)

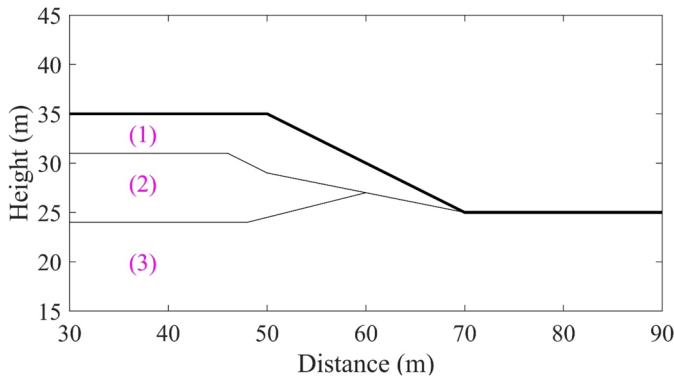


Fig. 9 Cross-section of Example 2

Table 3 Soil properties and statistics in Example 2

Item	Layer 1			Layer 2			Layer 3		
	γ (kN/m ³)	φ (°)	c (kPa)	γ (kN/m ³)	φ (°)	c (kPa)	γ (kN/m ³)	φ (°)	c (kPa)
Mean	19.5	38	—	19.5	23	5.3	19.5	20	7.2
COV	—	—	—	—	0.2	0.3	—	0.2	0.3
Distribution	—	—	—	—	Normal	Normal	—	Normal	Normal

3.2 Slope Example 2

Figure 9 presents the cross-section for the second example, which was examined in various previous studies (Ji and Low 2012; Kang et al. 2015). The soil properties are taken from earlier works, as tabled in Table 3. The soil stratification in the second slope is more complicated than in the first example. The basic MCS using 20,000 samples estimates a failure probability of 1.40%, as reported in Ji and Low (2012).

In this work, 4000 potential failure surfaces are analyzed using EET, resulting in a safety factor of 1.405, as shown in Fig. 10a. The performance of the PSO with 20 particles over 30 iterations is presented in Fig. 10b. The figure demonstrates that the optimization approach outperforms the EET by achieving the same safety factors after evaluating only 600 failure surfaces. This is a significantly lower number compared to the traditional method. Additionally, the optimization surveys potential candidates across a more comprehensive search area than the conventional EET, as depicted in Fig. 10.

Figure 11 evaluates the performance of both FA and PSO, while Fig. 12 summarizes the performance of PSO with various combinations of particle numbers and maximum iterations for Example 2. The results in both figures exhibit similar patterns to those observed in Example 1. Namely, the number of failure surfaces evaluated in FA varies across trial runs, whereas it remains consistent for each run using PSO. Additionally, FA requires a significantly larger number of surfaces to be investigated (with a mean of 2565) to identify the most critical surface. Although the

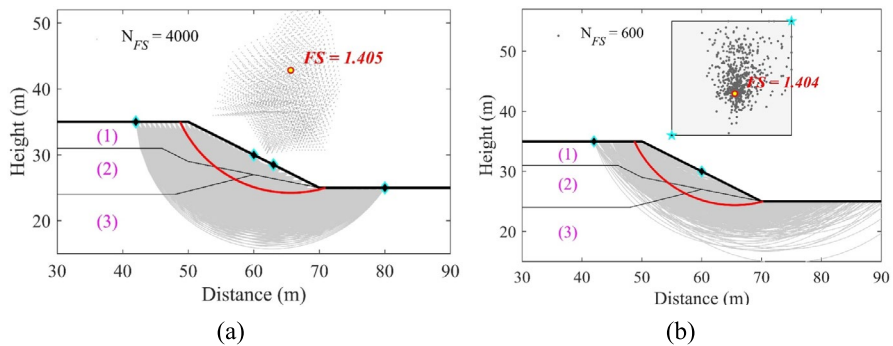


Fig. 10 Deterministic results of Example 2: **a** using EET and **b** using the optimization approach

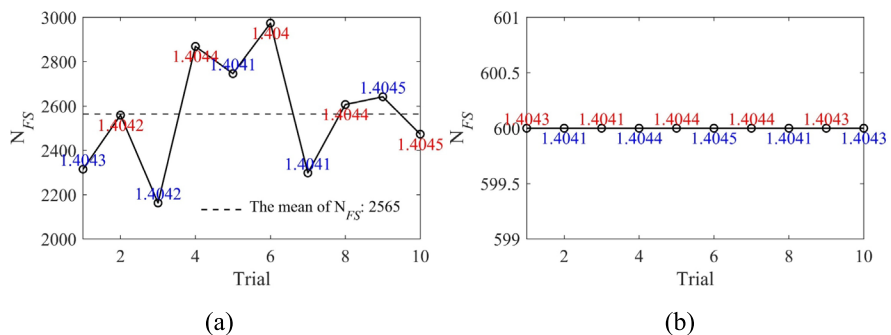
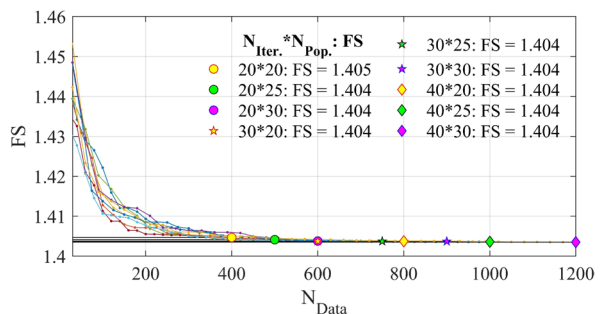


Fig. 11 Numbers of the failure surfaces required in the optimizations for Example 2: **a** using FA and **b** using PSO

Fig. 12 Comparisons for various combinations of particle number and iteration for Example 2



number of candidate surfaces needed by FA is less than that required by EET, it is still four times greater than that needed by PSO (using 30 iterations for 20 particles). Therefore, PSO is chosen again for the probabilistic analysis in Example 2 to minimize computational efforts.

The results of the MCSs using 50,000 samples, along with the EET and the optimization process, are shown in Table 4 for Example 2. This table also includes

Table 4 Comparisons of probabilistic results for Example 2

Method	P_f (%)	COV_p (%)	Number of failure surfaces in MCS	Reference
EET-based for critical failure surface and MCS (50,000 samples)	1.41	3.80	2×10^8 (50,000 $\times$ 4000)	This study
PSO-based for critical failure surface and MCS (50,000 samples)	1.45	3.70	3×10^7 (50,000 $\times$ 600)	This study
Basic MCS with 20,000 samples	1.40	5.98	–	Ji and Low (2012)

results from the literature for comparison. It shows that the PSO-based MCS provides better solutions than the EET. Notably, the optimization approach not only gives the best results but also reduces computational effort significantly. The fourth column of Table 4 indicates the number of failure surfaces considered in the MCSs, highlighting that optimization methods can lower computational demands by more than six times compared to the traditional EET method. Notably, the most critical failure surface varies with changes in soil properties, and the proposed optimization method quickly and accurately captures the critical failure mode for each MCS sample. As a result, the estimated probability of failure aligns closely with the system value.

3.3 Slope Example 3

In the previous two examples, soil properties are simulated as random variables. In this example, spatial variability is considered, so the soil properties are modeled as random fields. The proposed procedure is applied to this example to evaluate its performance under soil spatial variability. Incorporating spatial variability in soil properties considerably increases the complexity of the analysis by elevating its dimensionality. For example, conducting a single MCS with 10,000 simulations may require several hours or even days (Liu and Cheng 2016), whereas a MCS with 100,000 simulations could take 1.5 h per failure surface (Xiao et al. 2017), potentially extending the total computation time to several weeks to obtain the system failure probability. Therefore, developing more efficient methods that can replace conventional MCS for determining benchmark failure probabilities is crucial.

The slope illustrated in Fig. 13, referred to as the Congress Street cut and analyzed in previous studies (Ji and Low 2012; Li et al. 2016), is reassessed in this subsection. The statistical properties of the soils are taken from earlier works and tabled in Table 5. The slope comprises a sand and three clay layers situated on a hard stratum, as shown in Fig. 13a. In (Ji and Low 2012), soil properties were treated as random variables, whereas (Ding et al. 2023) simulated them as random fields. Due to soil formation processes and stress history, the horizontal spatial variability has a minor effect compared to the vertical variability. Therefore, it can be omitted to reduce complexity without significantly impacting accuracy (Wang et al. 2011). Additionally, since soil unit weights exhibit minimal variation with depth, they are

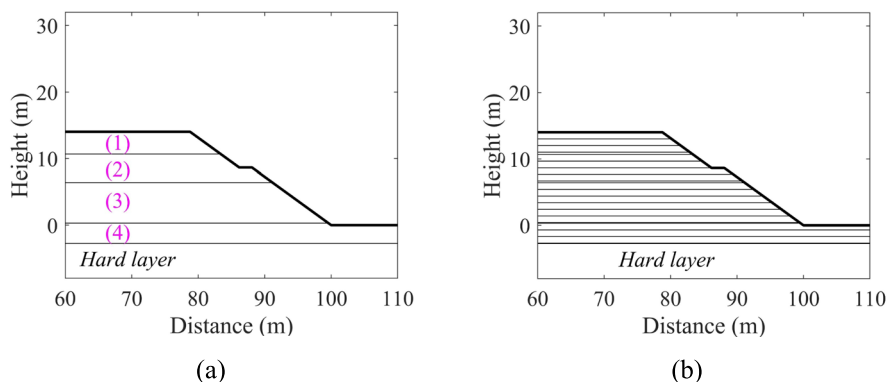


Fig. 13 Cross-section of Example 3

Table 5 Soil properties and statistics in Example 3

Soil	γ (kN/m ³)	φ (°)			c (kPa)		
		Mean	COV	Distribution	Mean	COV	Distribution
Sand	21	30	—	—	0	—	—
Clay 1	19.5	5	0.2	Lognormal	55	0.37	Lognormal
Clay 2	19.5	7	0.21	Lognormal	43	0.19	Lognormal
Clay 3	20	15	0.24	Lognormal	56	0.20	Lognormal

typically treated as random variables. In contrast, cohesion and friction angle are modeled as random fields when spatial variability is considered.

In this example, a vertical fluctuation scale of 6 m, based on previous work (Ding et al. 2023), is used to model the random fields for both soil cohesion and friction angle with depth. To simulate random fields, the soil profile is divided into 1-m layers (Fig. 13b). The spatial variability of soil properties with depth is then modeled using homogeneous lognormal random fields. The choice of autocorrelation function used to simulate random fields does not significantly impact the results, with the exponential correlation function being commonly recommended (Wang et al. 2011). Considering spatial variability in two directions, a failure probability of 0.27% was reported from a direct MCS with a sample size of 70,000 (Ding et al. 2023). In this work, a basic MCS with a size of 100,000 is conducted, and a failure probability of 0.264% is obtained. This value serves as the benchmark for this example when considering spatial variability in soil depth.

In deterministic calculations, a safety factor of 1.392 was reported in Li et al. (2016). In this work, safety factors of 1.398, 1.395, and 1.394 were determined using 8,000, 12,500, and 20,480 failure surfaces, respectively. It appears that 12,500 failure surfaces is an appropriate set for EET in this example (Fig. 14a). Investigations on the number of maximum iterations and the populations used in PSO will be presented later. Using 25 particles and 50 iterations for PSO, the most critical safety

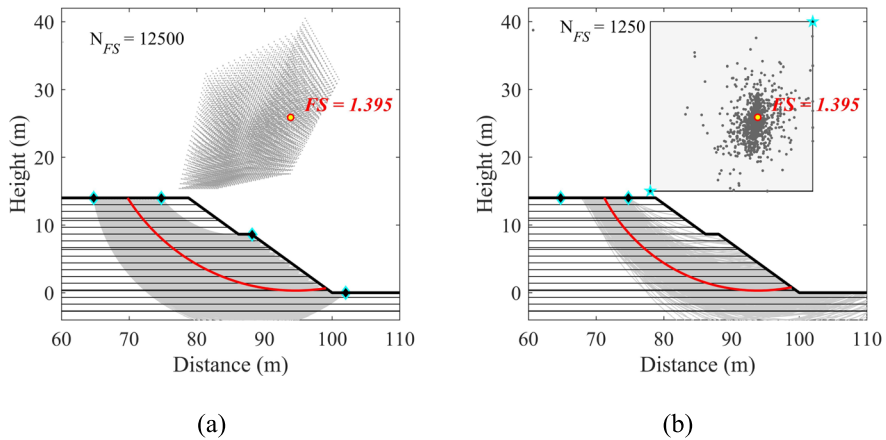


Fig. 14 Deterministic results of Example 3: **a** using EET and **b** using the optimization approach

factor of 1.395 is also determined after surveying 1,250 failure surfaces (50 iterations $\times$ 25 particles), as shown in Fig. 14b. Figure 14 demonstrates that employing PSO can reduce the number of surveyed surfaces by a factor of ten, leading to a corresponding tenfold decrease in computing time compared to the EET. Comparisons between the performances of PSO and FA for the slope stability assessment are made in Fig. 15 for Example 3. As observed in the previous examples, the number of candidate surfaces identified using FA is random in the figure. In contrast, the same number of candidate surfaces is systematically investigated using PSO. Although the number of candidate surfaces using FA (averaging 6214 over ten runs) is about half that of conventional EET (12,500), it is still significantly higher—approximately five times more—than that generated by PSO. Thus, the PSO is selected for further probabilistic analyses of Example 3.

Figure 16 compares various combinations of the numbers of particles and the maximum iterations for PSO. The figure indicates that examining fewer than 800 potential failure surfaces can not yield an accurate failure surface (FS). Moreover, Fig. 16 shows that 50 iterations are sufficient for 25 particles to obtain a good

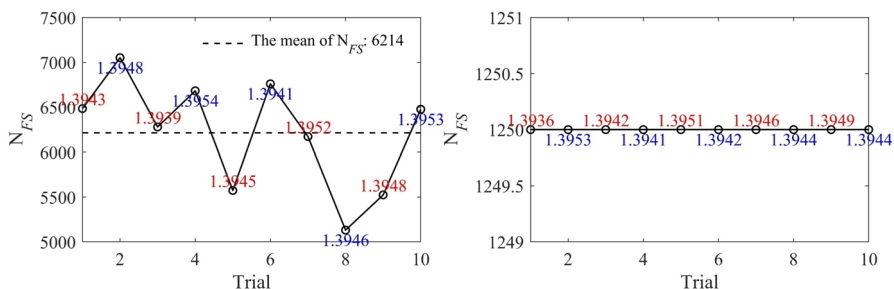


Fig. 15 Comparisons for various combinations of particle number and iteration for Example 3

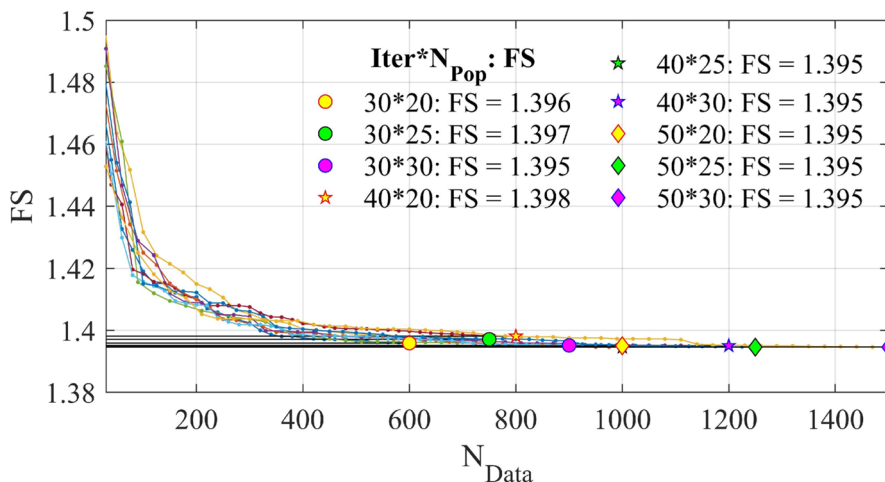


Fig. 16 Comparisons for various combinations of particle number and iteration for Example 3

solution, while the accuracy is not essentially enhanced when increasing the number of particles. This combination is applied to the optimization-based MCS for Example 3.

Figure 17 presents the correlations between elements at different depths, derived from the exponential correlation function. These correlations are utilized to simulate random fields for both cohesion and friction angles. Notably, the distances between elements are calculated using the center points of the elements at each depth. The figure illustrates that correlations decrease as the distance between elements increases. Figure 18 compares the cohesion values at a depth of 5.91 m with those at depths of 4.91 m and 0.36 m, all within Clay 2. Statistical information for the realizations is provided in the axis labels, and correlation coefficients are also reported in

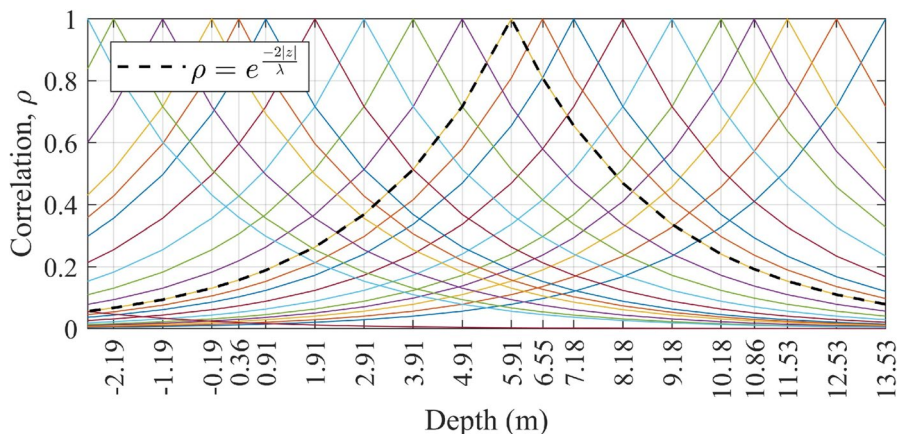


Fig. 17 Correlation in depth to simulate random fields for Example 3

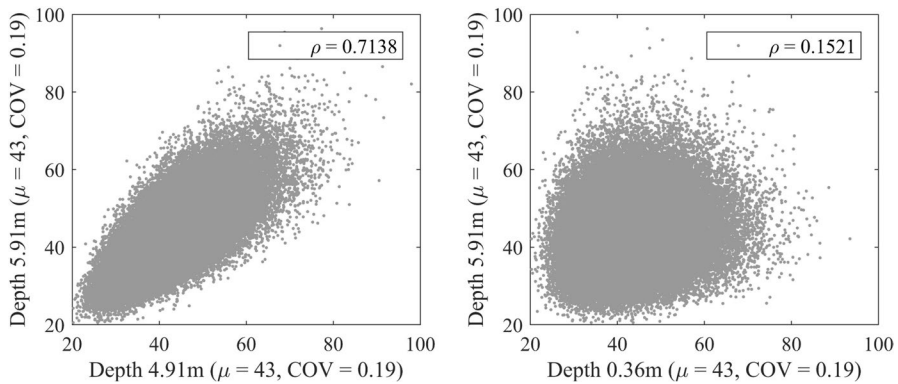


Fig. 18 Illustrative simulations of random fields for cohesions in Clay 2

Fig. 18. The statistical properties in Fig. 18 match those in Table 5, and the correlation coefficients align with those shown in Fig. 17 (the dashed lines). These results confirm the accuracy of the random field simulations.

With the random fields established, MCSs are conducted, and the results are presented in Table 6 for the two approaches used to generate potential failure surfaces. As with the previous examples, the number of possible failure surfaces considered is also reported in the fourth column of the table. Additionally, Table 6 includes MCS results from the literature for comparison (considering spatial variability in two directions). Table 6 indicates that spatial variability in the horizontal direction has a minimal effect. Furthermore, the results obtained using the two methods in this study are consistent. However, the proposed optimization method significantly reduces the number of safety factor estimations required in MCSs by more than ten times compared to the EET, as shown in the fourth column of Table 6. For example, while a basic MCS may require 10 h to complete, the proposed method achieves the same results in just one hour.

4 Summary and Conclusions

This study proposed an efficient optimization approach combined with the MCS for assessing slope reliability. The proposed optimization helps mitigate the number of failure surfaces involved in slope stability without limiting the investigation domains. Subsequently, this approach can be seamlessly integrated with standard MCS to estimate failure probabilities accurately and effectively. Notably, the most critical failure surface may vary across samples, and the proposed optimization effectively identifies these essential failure modes with precision. Two optimization algorithms are employed, and three illustrative examples are analyzed. The key findings are summarized as follows:

Movements of fireflies in FA are adjusted by referring to the positions of other fireflies. That means FA considers the interactions between the fireflies during

Table 6 Comparisons of probabilistic results for Example 3

Method	P_f (%)	COV_p (%)	Number of failure surfaces in MCS	Reference
EET-based for critical failure surface and MCS (100,000 samples)	0.264	6.2	12.5×10^8 (100,000 $\times$ 12,500)	This study
PSO-based for critical failure surface and MCS (100,000 samples)	0.264	6.2	12.5×10^7 (100,000 $\times$ 1250)	This study
Basic MCS with 70,000 samples (considering both horizontal and vertical directions)	0.270	7.3	–	Ding et al. 2023)

the search for the objective; hence, it requires more computing effort, as depicted by the large number of candidate failure surfaces in slope stability analyses. Alternatively, PSO is more suitable for slope stability problems, as it requires significantly fewer surfaces to be explored when identifying the most critical failure surface. Across the three illustrative examples, the FA required examining about six times the number of failure surfaces compared to PSO to achieve the same level of accuracy. The most critical center tends to concentrate on specific regions. Therefore, particles of PSO focus on these potential zones to refine the search process. In contrast, the EET requires an equivalent calculation of all possible points to identify the most critical one. As a result, the proposed optimization approach can accurately and efficiently pinpoint the most critical failure surface across a broader range of potential surfaces without increasing computational effort. The results obtained using PSO are at least comparable, and often superior, to those achieved with the traditional EET, as demonstrated in Example 1.

Using PSO can significantly reduce the number of surfaces needed to identify the most critical mode compared to conventional EET. For example, the number of surfaces surveyed can be reduced by at least fivefolds, as demonstrated in Example 1, and by up to tenfold, as seen in Example 3. Since spatial variability can extend analysis time to weeks, the proposed approach can shorten computing time by up to ten times, greatly enhancing efficiency. Therefore, integrating PSO with MCS can significantly accelerate computational processes while preserving the accuracy of conventional methods.

This study provides an efficient alternative for establishing benchmark failure probabilities by applying MCSs to the original implicit slope stability problems, with the accuracy validated against the results from basic MCS. While BSM is employed in this work, the optimization method can be extended to other slice techniques. Moreover, PSO is expected to outperform FA due to its more effective particle interactions, making PSO a recommended option for improving MCS computation time.

Author Contribution Nhu Son Doan: conceptualization, methodology, software, validation, writing—original draft, and writing—review and editing.

Funding This study was supported by Vietnam Maritime University.

Data Availability Data will be made available on reasonable request.

Declarations

Ethics Approval and Consent to Participate Not applicable.

Consent for Publication Not applicable.

Competing Interests The authors declare no competing interests.

References

- AASHTO.: LRFD Bridge Design Specifications. 8th ed. AASHTO: Washington, DC, USA. (2017)
- Ahmad, F., Samui, P., Mishra, S.S.: Probabilistic slope stability analysis on a heavy-duty freight corridor using a soft computing technique. *Transp. Infrastruct. Geotechnol.* **11**, 2090–2113 (2024). <https://doi.org/10.1007/s40515-023-00365-4>
- Banks, A., Vincent, J., Anyakoha, C.: A review of particle swarm optimization. Part I: Background and development. *Nat. Comput.* **6**:467–84. (2007). <https://doi.org/10.1007/s11047-007-9049-5>
- Boudiaf, K., Benmeddour, D., Baazouzi, M., Mabrouki, A., Mellas, M.: A Numerical investigation of the effect of isotropic spatially variable tensile strength on slope stability. *Transp. Infrastruct. Geotechnol.* **6**, 268–288 (2019). <https://doi.org/10.1007/s40515-019-00080-z>
- Ching, J., Phoon, K.-K., Hu, Y.-G.: Efficient evaluation of reliability for slopes with circular slip surfaces using importance sampling. *J. Geotech. Geoenviron. Eng.* **135**, 768–777 (2009). [https://doi.org/10.1061/\(asce\)gt.1943-5606.0000035](https://doi.org/10.1061/(asce)gt.1943-5606.0000035)
- Ding, J., Zhou, J., Cai, W.: An efficient variable selection-based Kriging model method for the reliability analysis of slopes with spatially variable soils. *Reliab. Eng. Syst. Saf.* **235**, 109234 (2023). <https://doi.org/10.1016/j.ress.2023.109234>
- Dinh, H.-B., Mac, V.H., Doan, N.S.: Comparative study on semi-probabilistic design methods to calibrate load and resistance factors for sliding stability design of caisson breakwaters. *Ocean. Eng.* **293**, 116573 (2024). <https://doi.org/10.1016/j.oceaneng.2023.116573>
- Djaneye-Boundjou, O., Ordóñez, R., Gazi, V.: Adaptive particle swarm optimization. *Int. Conf. Control. Autom. Syst.* **39**, 440–445 (2009). <https://doi.org/10.1109/ICCAS.2013.6703971>
- Doan, N.S.: Reliability analysis and uncertainty quantification of clay and sand slopes stability evaluated by Fellenius and Bishop's simplified methods. *Int. J. Geo-Eng.* **14**. (2023). <https://doi.org/10.1186/s40703-023-00200-2>
- Doan, N.S., Mac, V.H., Dinh, H.: Machine learning applications to load and resistance factors calibration for stability design of caisson breakwater foundations. *Comput. Geotech.* **169**, 106225 (2024a). <https://doi.org/10.1016/j.compgeo.2024.106225>
- Doan, N.S., Mac, V.H., Dinh, H.: An optimization-based method of calibrating load and resistance factors: application to slope and breakwaters' foundation stability. *Ocean. Eng.* **313**, 119409 (2024b). <https://doi.org/10.1016/j.oceaneng.2024.119409>
- Duncan, J.M., Wright, S.G., Brandon, T.L.: Soil strength and slope stability, 2nd edn. John Wiley & Sons Inc, Hoboken, New Jersey, USA (2014)
- Haldrar, A., Mahadevan, S.: Probability, reliability and statistical methods in engineering design. John Wiley, Hoboken, NY, USA (2000)
- Javankhosdel, S., Cami, B., Chenari, R.J., Dastpak, P.: Probabilistic analysis of slopes with linearly increasing undrained shear strength using RLEM approach. *Transp. Infrastruct. Geotechnol.* **8**, 114–141 (2021). <https://doi.org/10.1007/s40515-020-00118-7>
- Ji, J., Low, B.K.: Stratified response surfaces for system probabilistic evaluation of slopes. *J. Geotech. Geoenviron. Eng.* **138**, 1398–1406 (2012). [https://doi.org/10.1061/\(asce\)gt.1943-5606.0000711](https://doi.org/10.1061/(asce)gt.1943-5606.0000711)
- Kang, F., Han, S., Salgado, R., Li, J.: System probabilistic stability analysis of soil slopes using Gaussian process regression with Latin hypercube sampling. *Comput. Geotech.* **63**, 13–25 (2015). <https://doi.org/10.1016/j.compgeo.2014.08.010>
- Li, D.Q., Zheng, D., Cao, Z.J., Tang, X.S., Phoon, K.K.: Response surface methods for slope reliability analysis: review and comparison. *Eng. Geol.* **203**, 3–14 (2016). <https://doi.org/10.1016/j.enggeo.2015.09.003>
- Li, J., Wei, X., Li, B., Zeng, Z.: A survey on firefly algorithms. *Neurocomputing* **500**, 662–678 (2022). <https://doi.org/10.1016/j.neucom.2022.05.100>
- Liu, L.L., Cheng, Y.M.: Efficient system reliability analysis of soil slopes using multivariate adaptive regression splines-based Monte Carlo simulation. *Comput. Geotech.* **79**, 41–54 (2016). <https://doi.org/10.1016/j.compgeo.2016.05.001>
- MLIT.: Technical standard and commentaries for port and harbor facilities in Japan. Tokyo, Japan: Ministry of Land, Infrastructure, Transport and Tourism (MLIT): Ports and Harbors Bureau. (2020)
- Rukhaiyar, S., Alam, M.N., Samadhiya, N.K.: A PSO-ANN hybrid model for predicting factor of safety of slope. *Int. J. Geotech. Eng.* **12**, 556–566 (2018). <https://doi.org/10.1080/19386362.2017.1305652>

- Van, D.P., Huh, J., Doan, N.S., Kwak, K.: Influence of spatial variability of soil strength on load and resistance factors calibration for the design of breakwater foundation. *Ocean. Eng.* **268**, 113441 (2023). <https://doi.org/10.1016/j.oceaneng.2022.113441>
- Wang, Y., Cao, Z., Au, S.K.: Practical reliability analysis of slope stability by advanced Monte Carlo simulations in a spreadsheet. *Can. Geotech. J.* **48**, 162–172 (2011). <https://doi.org/10.1139/T10-044>
- Xiao, T., Li, D.Q., Cao, Z.J., Tang, X.S.: Full probabilistic design of slopes in spatially variable soils using simplified reliability analysis method. *Georisk* **11**, 146–159 (2017). <https://doi.org/10.1080/17499518.2016.1250279>
- Yang X.S. Firefly algorithms for multimodal optimization. *Lect Notes Comput Sci (Including Subser Lect Notes Artif Intell Lect Notes Bioinformatics)*. 5792 LNCS:169–78. (2009). https://doi.org/10.1007/978-3-642-04944-6_14
- Yang, X.: Metaheuristic Optimization : algorithm analysis. *Proc 10th Int Symp Exp Algorithms (SEA 2011)*. 6630:21–32. (2011). https://doi.org/10.1007/978-3-642-20662-7_2
- Yang, Z.Y., Li, D.Q., Cao, Z.J., Tang, X.S.: System reliability of soil slope using generalized subset simulation. *Yantu Lixue/Rock Soil Mech.* 39:1–50. (2018). <https://doi.org/10.16285/j.rsm.2016.0780>
- Yi, P., Wei, K., Kong, X., Zhu, Z.: Cumulative PSO-Kriging model for slope reliability analysis. *Probabilistic Eng Mech* **39**, 39–45 (2015). <https://doi.org/10.1016/j.probengmech.2014.12.001>
- Zhang, J., Huang, H.W., Juang, C.H., Li, D.Q.: Extension of Hassan and Wolff method for system reliability analysis of soil slopes. *Eng. Geol.* **160**, 81–88 (2013). <https://doi.org/10.1016/j.enggeo.2013.03.029>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.